



Cradle Vibes Baby Soother

Engineering discussion brief

1. Project goal

The goal is to develop a baby soother that delivers reliable, natural-feeling motion with significantly improved battery performance, while remaining simple and intuitive for parents to use across multiple environments (crib, stroller, pack-n-play).

2. What we know from research

From reviewing the category, a few consistent themes emerged:

- Battery life is the #1 complaint across products
- Attachment is frustrating across different use cases
- Motor noise level can disrupt sleep

3. Core features (priority areas)

These are the features we see as central to the product experience:

Rocking motion

- Core function of the product
- Frida can be used as a baseline reference
- Opportunity to explore more natural-feeling motion vs purely mechanical vibration
- Open to different motion profiles (continuous, pulsed, etc.)

Cry Detection + Operating Modes (priority feature)

- Product should explore two user-selectable operating modes:
 1. Cry Detection Mode: automatic activation when crying is detected
 2. Continuous Mode: user manually activates soothing cycle
- Interested in exploring the most reliable and efficient approach

Sound (secondary feature)

- Basic lullabies / soothing sounds
- Volume roughly in line with VTech Hush or similar
- Not a primary differentiator, but useful if implemented well

Multi-environment use

- Should work across stroller, bassinet, crib, and pack-n-play
- We see this as a key part of the product's value

Baseline performance targets

- Approx. 60-minute soothing cycle (rocking & playing music)
- Battery performance at least on par with Frida as a starting point

4. Primary design opportunity: extended runtime

One of the biggest gaps in the category is limited runtime. We see extended soothing (rocking) duration as a key opportunity to differentiate this product.

We understand this will involve trade-offs between battery size, product size/weight, and motor performance, and we're open to exploring different approaches.

Some directions we'd like to discuss:

- Plug-in continuous mode for home use (crib / pack-n-play)
- Extended or modular battery strategies
- Firmware strategies to reduce power consumption
- Smarter motion cycles (intermittent vs continuous use)

If we had to prioritize, improving runtime and delivering reliable motion are more important than adding additional features.



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5. Attachment (key experience area)

Attachment is a major pain point in the category and an important part of the user experience.

We'd like to develop a solution that:

- Works across multiple environments (stroller, crib, pack-n-play, bassinet)
- Is quick and intuitive to attach and remove
- Feels secure and stable during use

We see this as an area where thoughtful design can significantly improve the product. David will focus on designing solutions for this experience; however, all ideas are welcomed.

6. Non-negotiable

- Must be safe, durable, and appropriate for baby products.
- Must be simple and intuitive to use (especially during nighttime use)
- Must be water-resistant (drool, light moisture, general cleaning)
- Should operate quietly enough to not disrupt sleep

Collaboration Approach

This is intended as a starting point for discussion. We expect there to be trade-offs between performance, size, battery life, and cost, and we're looking forward to working collaboratively to define the right balance.

Following our initial discussion, we can refine this into a more detailed and formal feature specification.